

SNAKE GAME USING PYTHON

Project Statement

The objective of this project is to develop a simple and interactive **Snake Game using Python and the Pygame library**. The game allows the player to control a snake using keyboard keys and guide it towards randomly generated food.

Whenever the snake successfully eats the food, its length and score increase. The game includes proper movement controls, random food generation, score tracking, snake growth, collision detection, and increasing difficulty as the game progresses.

The game ends when the snake collides with the boundary of the game screen or with its own body. A **Game Over** screen is displayed after the collision, providing options to restart the game or quit it.

The project also includes **graphics and sound effects** to make the game more interactive and enjoyable. The main purpose of this project is to understand and apply Python programming concepts such as variables, loops, conditional statements, functions, lists, event handling, collision detection, and the use of external libraries such as Pygame.

Objectives

- To develop a functional Snake Game using Python.
- To understand the basic concepts of game development.
- To implement keyboard-based player controls.
- To generate food randomly on the game screen.
- To increase the snake's length and score when food is collected.
- To implement collision detection.
- To create a Game Over and restart system.
- To use graphics and sound effects using Pygame.
- To improve programming and problem-solving skills.

Technologies Used

- **Programming Language:** Python
- **Library:** Pygame
- **Development Environment:** Visual Studio Code
- **Concepts:** Loops, Functions, Lists, Conditions, Events and Collision Detection

Expected Outcome

The final application will be a playable Snake Game in which the player controls the snake, collects food, increases the score, and tries to achieve the highest possible score without

colliding with the wall or itself.